AI-augmented curation of drug vocabulary for knowledge-graph–driven repurposing and analytics

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Abstract

Automated text and data mining of relevant scientific content facilitates drug repurposing and the discovery of unexplored target–indication pairs by enabling comprehensive and precise mappings between biomedical entities. For this purpose, Elsevier SciBite curates a NER-focused drug vocabulary grounded in ChEMBL, an open-source, curated database of drug-like molecules. This vocabulary encompasses >400,000 compounds, curated and enriched with synonyms, brand names, and research codes to enhance coverage and disambiguation.

The vocabulary integrates critical drug-related metadata, such as mechanism of action, target and indication profiles that are essential for accurate ontology modeling and semantic reasoning. This rich metadata supports construction of drug modelling, and downstream detailed, multi-relational knowledge graphs that enable nuanced queries and advanced analytics. The value of the vocabulary depends heavily on maintaining up-to-date and accurate coverage of drug entities and their relationships. Manual curation, while essential for quality, is costly and time-consuming, raising the question of how much curation is sufficient to keep pace with the rapidly evolving biomedical domain. To address this challenge, we are bringing in automated techniques including a generative AI (genAI) methodology to generate synonyms and assess domain relevance. AI-driven suggestions augment the work of SMEs, who validate and refine annotations, striking a balance between automation and expert oversight. Combining genAI, curated vocabularies aligned with regulatory standards, rich metadata, and SME expertise—yields a robust, scalable framework for biomedical entity mapping. The resulting datasets empower knowledge graph applications and downstream analytics, enhancing discoverability, data integration, and insight generation in drug repurposing and biomedical research.

Keywords

Semantic Enrichment, Drug modelling, GenAI

SWAT4HCLS 2026

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